

MUHAMMAD BASITH K

AI/ML Engineer — Machine Learning — Computer Vision

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Professional Summary

Computer Science Engineer with hands-on experience in machine learning, deep learning, computer vision, and AI-assisted application development. Research experience includes uncertainty-aware satellite image segmentation using SegFormer and Evidential Deep Learning (EDL), spanning semantic segmentation, epistemic uncertainty estimation, out-of-distribution detection, and GPU-based experimentation. Presented research at ICIET-2026 and co-authored a conference paper on uncertainty quantification for satellite image segmentation.

Technical Skills

Languages: Python, C++, Java, SQL

Machine Learning: Deep Learning, Computer Vision, Semantic Segmentation, Model Evaluation, Uncertainty Quantification, Out-of-Distribution Detection, Feature Engineering, Data Preprocessing

Deep Learning & AI: PyTorch, SegFormer, Vision Transformers, Evidential Deep Learning, Epistemic Uncertainty, MC Dropout, CUDA, Google Gemini, LLM Integration

ML/Data Libraries: NumPy, Pandas, Rasterio, Plotly, Albumentations

Engineering: Git, GitHub, OOP, Data Structures & Algorithms, Debugging, Agile/Scrum

Backend (applied to AI-enabled apps): Django, Django REST Framework, PostgreSQL, REST APIs

Research Experience

Uncertainty-Aware SegFormer for Satellite Image Segmentation

GitHub

Computer Vision — Remote Sensing — Evidential Deep Learning

- Developed an uncertainty-aware semantic segmentation framework combining the SegFormer MiT-b3 architecture with Evidential Deep Learning (EDL) to estimate epistemic uncertainty from a single forward pass.
- Trained and evaluated the framework on the DeepGlobe Land Cover dataset, achieving **65.81% mIoU** for satellite image semantic segmentation.
- Investigated the relationship between predictive uncertainty and segmentation errors, achieving an uncertainty-error correlation of $\rho = 0.506$.
- Compared single-pass EDL against MC Dropout across accuracy, uncertainty estimation, calibration, and inference efficiency, with EDL inference measured at **12.9s**.
- Evaluated out-of-distribution robustness using synthetic image corruptions, obtaining a Spearman correlation of $\rho = 0.686$ for OOD uncertainty behavior.

Publication & Conference Presentation

Uncertainty-Aware SegFormer for Satellite Image Segmentation Using Evidential Deep Learning

Shiju Kumar P S, Hrishikesh S Raj, Muhammad Basith K

- Conference paper presented at the **Second International Conference on Innovations in Engineering & Technology (ICIET-2026)**, Ilahia College of Engineering & Technology, Muvattupuzha, Kerala, India, **August 20–21, 2026**, focused on integrating Evidential Deep Learning with SegFormer MiT-b3 for uncertainty-aware satellite image segmentation, covering epistemic uncertainty, calibration, inference efficiency, and OOD detection.

AI/ML Projects

TerraWatch – Satellite Image Analysis & Monitoring Platform

GitHub

Python, Machine Learning, Rasterio, NumPy, Pandas, Plotly

- Built a geospatial analytics pipeline processing six high-resolution raster datasets using Rasterio, NumPy, and Pandas for disaster-susceptibility mapping via clustering-based analysis.
- Improved usable spatial data coverage to $\sim 90\%$ through raster preprocessing and alignment; built interactive Plotly visualizations for exploring spatial risk regions.

Experience

Python Full Stack Developer Intern (Hybrid)

Jan 2026 – July 2026

Gen Corpus Data Hub, Kozhikode, Kerala

- Engineered and shipped 15 full-stack features using Python, Django, DRF, React.js, and PostgreSQL, implementing 24 RESTful API endpoints and resolving 43 bugs.
- Investigated failures across frontend, backend, API, and database boundaries, tracing root causes and implementing fixes – applying the same debugging and evaluation discipline later used in ML model analysis.

Education

SRM Institute of Science and Technology, Chennai, India

Aug 2022 – May 2026

B.Tech in Computer Science and Engineering

CGPA: 7.63/10.0

Certifications

AWS Academy – Machine Learning Foundations | Full Stack Open – University of Helsinki (5 ECTS, Grade 5) | Python Programming (Iskima, 90%)

Achievements

- Co-authored and presented research on uncertainty-aware satellite image segmentation using SegFormer and Evidential Deep Learning at ICIET-2026.
- Led a three-member research team working on uncertainty-aware semantic segmentation and uncertainty quantification for satellite imagery.
- Earned HackerRank 5-star badges in C, C++, and Java, along with the Problem Solving badge.